

Phase 1: Explainable AI (XAI) TalkToModel Replication — Summary & Analysis

Replication of “Explaining Machine Learning Models with Interactive Natural Language Conversations
Using TalkToModel” (Slack et al., Nature Machine Intelligence, 2023)

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1. Project Overview

TalkToModel democratizes Explainable AI (XAI) via an end-to-end dialog system: users converse with a model in natural language (e.g., “Why did you predict this for patient 5?”, “What if their age was younger?”) instead of writing code. Pipeline: (1) natural language input → (2) fine-tuned **T5 token parser** maps English to grammar tokens (e.g., `filter id 5 and explain features [e]`) → (3) **action mapping engine** runs the tokens as procedural logic on the data and model → (4) **XAI engine (MegaExplainer)** computes feature importances (e.g., SHAP) or statistics → (5) **template response** returns readable text.

2. Replication Objectives

- **Restore the legacy environment:** Python 3.10, Flask 2.0.3, NumPy 1.21.6, needed to load the authors’ pickled models.
- **Evaluate T5 parsing accuracy** and **validate explanation quality** (MegaExplainer/SHAP vs. a custom LIME explainer).
- **Test real-time feasibility** and **demonstrate interactivity** via CLI tools for the Diabetes, COMPAS, and German Credit datasets.

3. Key Findings and Discrepancies

Metric	Replication	Reference
T5-Small parsing accuracy (Diabetes)	97.33%	66.8% in paper (+30.53 pp)
Top-3 feature overlap (SHAP vs. LIME)	58.33%	—
Spearman rank correlation (SHAP vs. LIME)	0.364	—
Parse-to-answer latency (CPU only)	0.3–0.7 s	No GPU needed
User study: “Easier to use” / “Faster to answer”	86.2% / 86.2%	Matches paper
User study: “Prefer for future use”	69.0%	86.2% in paper

A. Parsing accuracy is split-dependent. The paper neither documents nor provides its held-out splits, so evaluating on the supplied CSV likely involved training overlap or an imbalanced distribution (cause unconfirmed). The paper’s T5-Base model is also missing from the Hugging Face hub and could not be verified.

B. Explanations are algorithm-dependent. Moderate SHAP–LIME agreement shows global trends exist but local attributions diverge (SHAP estimates marginal contributions; LIME fits a local surrogate), an important caveat for medical or criminal-justice use.

C. Real-time feasibility holds. Typical queries ran in 0.3–0.7 s on CPU alone, supporting the claim that the framework is lightweight.

D. User-study numbers differ. Raw responses (`ttm-user-study-responses.csv`) reproduce the 86.2% ease and speed claims but give 69.0% for future use, suggesting undocumented aggregation or omitted groups.

4. Conclusion

The replication supports the conceptual feasibility of TalkToModel: constrained T5 generation combined with rule-based execution over deterministic XAI logic is effective and fast. The end-to-end scripts (`07_end_to_end_talktomodel_diabetes.py`, `compas.py`, `german.py`) run robustly as interactive CLI tools. Limitations remain:

- **Brittle packaging:** pickled models tied to specific NumPy/Pandas/Flask versions age poorly.
- **Missing artifacts:** absent data splits and T5-Base weights prevent 1:1 validation of NLP accuracy claims.
- **Data discrepancies:** mismatched user-study figures underscore the need for rigorous, open data publishing.